MPKin-YSTR User Manuel

Version 1.0

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Introduction

MPKin-YSTR is a software program designed for kinship analysis with Y-STR haplotypes. This software was developed with the support from the National Institute of Justice (NIJ) 2020-DQ-BX-0018. There are three major functions implemented in this tool:

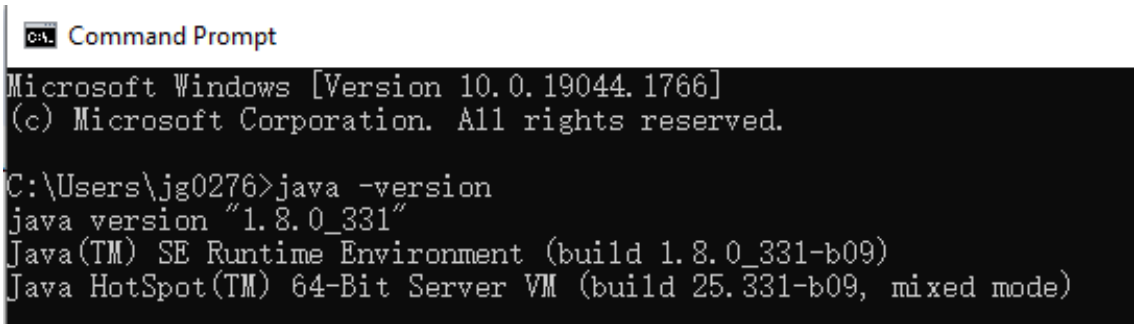
- Y-STR haplotype comparisons (i.e., 1:1, 1:N, and N:N comparisons)
- Pedigree likelihood ratio for kinship analysis with Y-STR profiles
- Draw mismatch distributions to decide a threshold for determining if two profiles are from the same lineage or not

Software Installation

Please make sure your computer (either Windows, Linux, or Mac) has the latest version of Java installed. Please go to https://www.java.com/ to download and install Java. To find out the version of Java installed on your computer or to find out if your computer has Java installed, please type the following command in the command window:

java -version

The following is an example for the Windows system.



Program execution

The program can be executed by double-clicking the "MPKin-YSTR.jar" file.

The program may also be executed by typing "java -jar MPKin-YSTR.jar" in the command windows. The approach is recommended if a bug is found, so that the error trace may be seen in the command windows.

Input files

There are two types of data files that may be needed in this software: Y-STR haplotype profiles and Y-STR mutation rates

Y-STR haplotype profiles

MPKin-YSTR accepts two formats for the Y-STR haplotype profiles: CMF format defined by FBI (version 3.2 or later) or Excel file (.xlsx).

Common Message Format (CMF) file

The CMF file is a specific XML file, which is defined by the FBI CODIS system. More details may be found by https://www.fbi.gov/file-repository/codis-rapid-import-cmf-interface-specification.pdf. An example CMF file is provided together with the software.

Excel file

The format of the Excel file that contains the Y-STR haplotypes is as follows:

1. The first row contains the "SampleName" and the marker names; one cell for one marker name
2. The followed rows contain the samples; one row for one sample;
3. The first column has to be the sample name
4. The followed columns contain the genotypes of the Y-STR markers; one cell in each column for one genotype of one specific sample
5. An empty cell means missing data

The following table is an example. An example Excel file is also provided with the software.

SampleName	DYS389 I	DYS389 II	DYS390
AA-1	13	28	22
AA-2	14	31	
AA-3	14	32	21

Y-STR mutation rates

The mutation rates file is a simple text file. The file contains two columns, and these two columns are separated by tab.

The first column contains the marker name. The marker name may contain space. The marker names with space and without space are different. For example, "DYS389 I" is different from "DYS389I". Please make sure the marker names in the mutation file and the haplotype file are consistent. Different software may export different Y-STR marker names, which may lead to errors in the subsequent analysis. Specifically, please check the space within the marker names, and check that "YGATAH4" is different from "GATAH4 ".

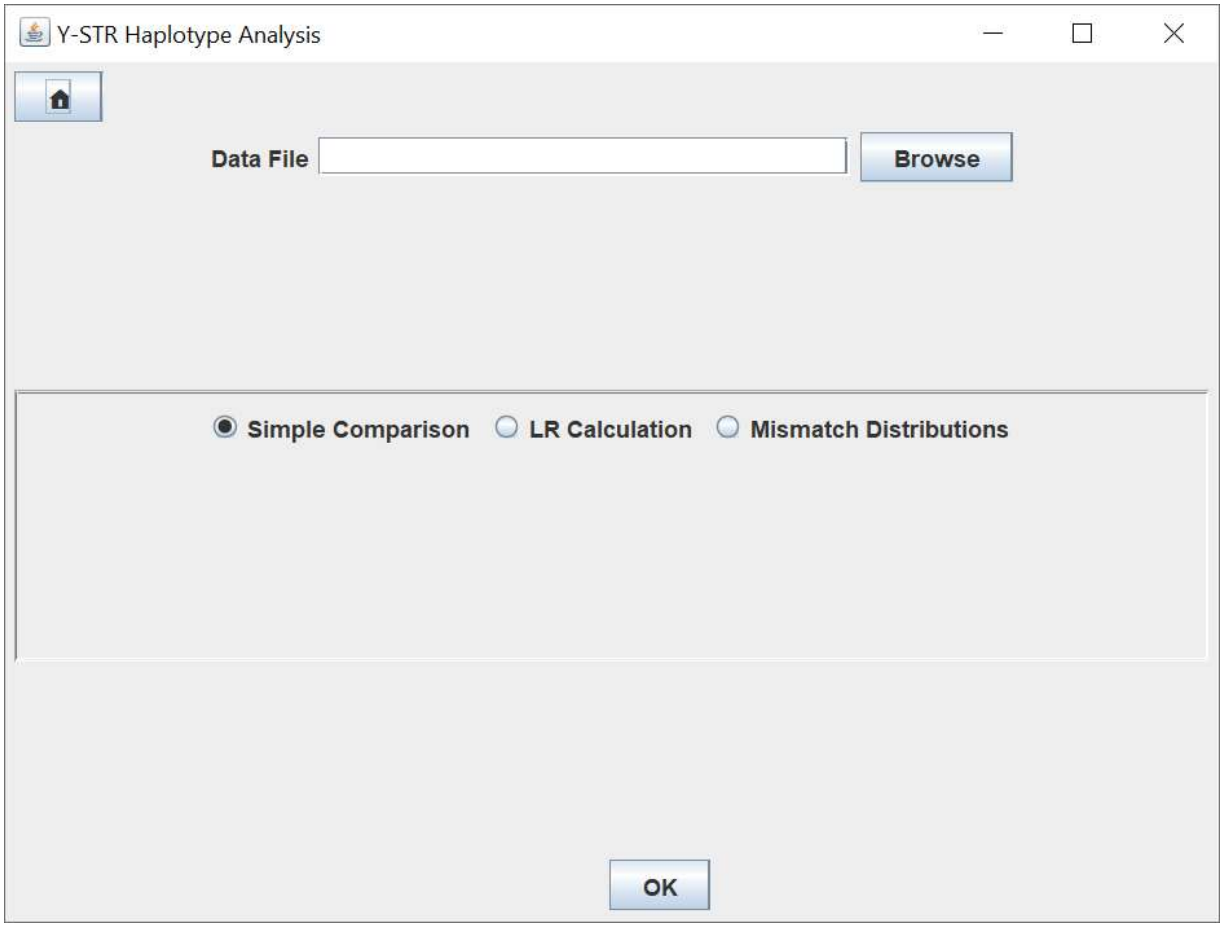
The second column contains the mutation rates of the Y-STR markers.

MPKin-YSTR has provided a mutation rate file "AllCommonYSTRMutationRates – nameWithSpace.txt". The mutation rates were obtained from YHRD (https://yhrd.org/pages/resources/mutation_rates) in March 2020.

Functions

MPKin-YSTR has three major functions: Simple Comparison, LR Calculation, and Mismatch Distributions.

First, users need to browse and upload a data file (i.e., Y-STR haplotype file). Second, select a function. Third, click “OK” button to move forward.

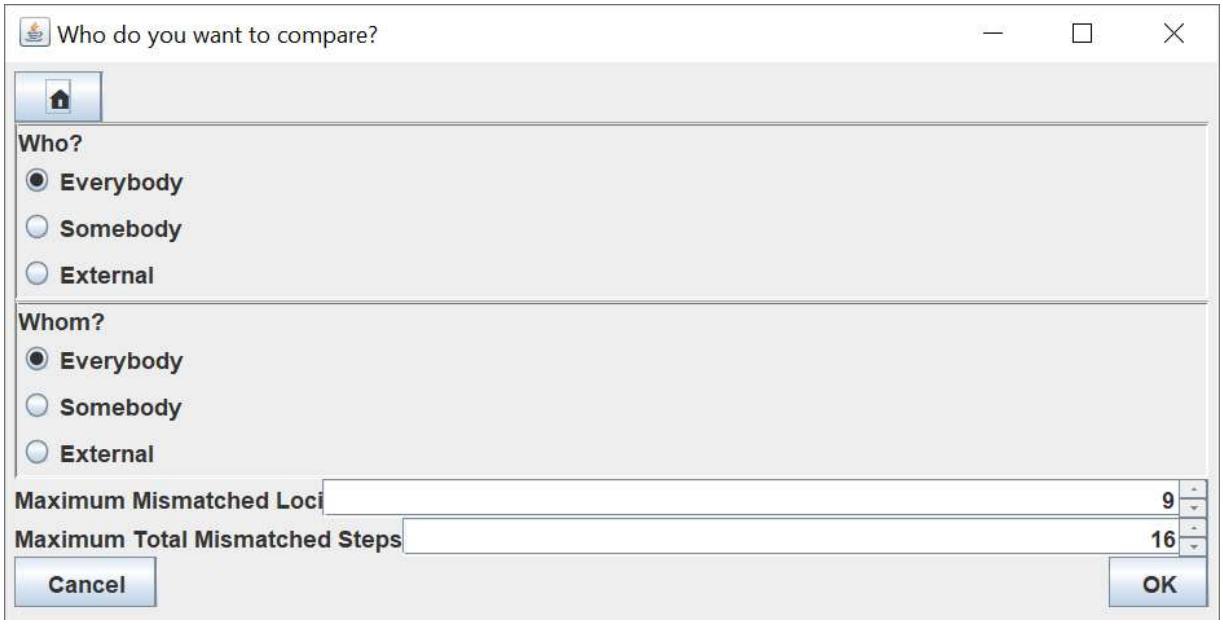


Profile Comparisons

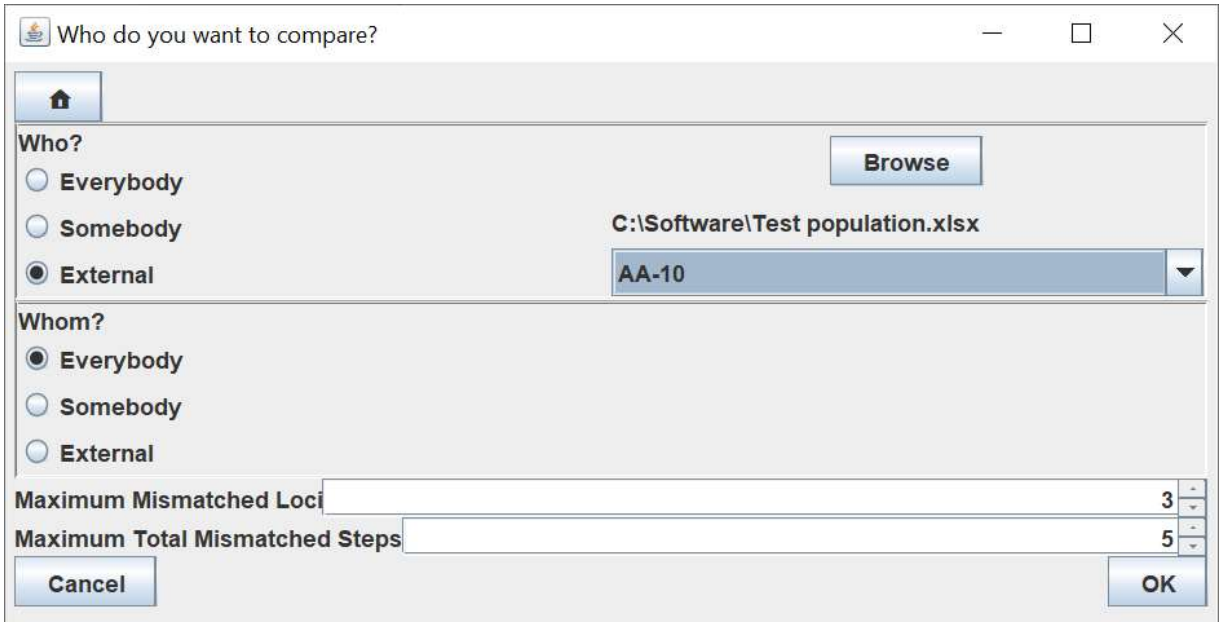
MPKin-YSTR allows comparing Y-STR profiles in 1:1 (i.e., one profile vs. another profile), 1:N (i.e., one profile vs. multiple profiles), and N:N (i.e., pairwise comparisons with a profile set).

1:1 → Somebody vs. Somebody
1:N → Everybody vs. Somebody, or vice verse
N:N → Everybody vs. Everybody

Two thresholds were defined to screen the profile comparison results: (1) the maximum number of mismatched loci (e.g., 3) and (2) the maximum number of mismatched steps (e.g., 5). The software only displays the profile pairs passing the screening thresholds.



Users may also upload an additional data profile by the “External” option. One profile in the additional uploaded file should be decided in comparison.

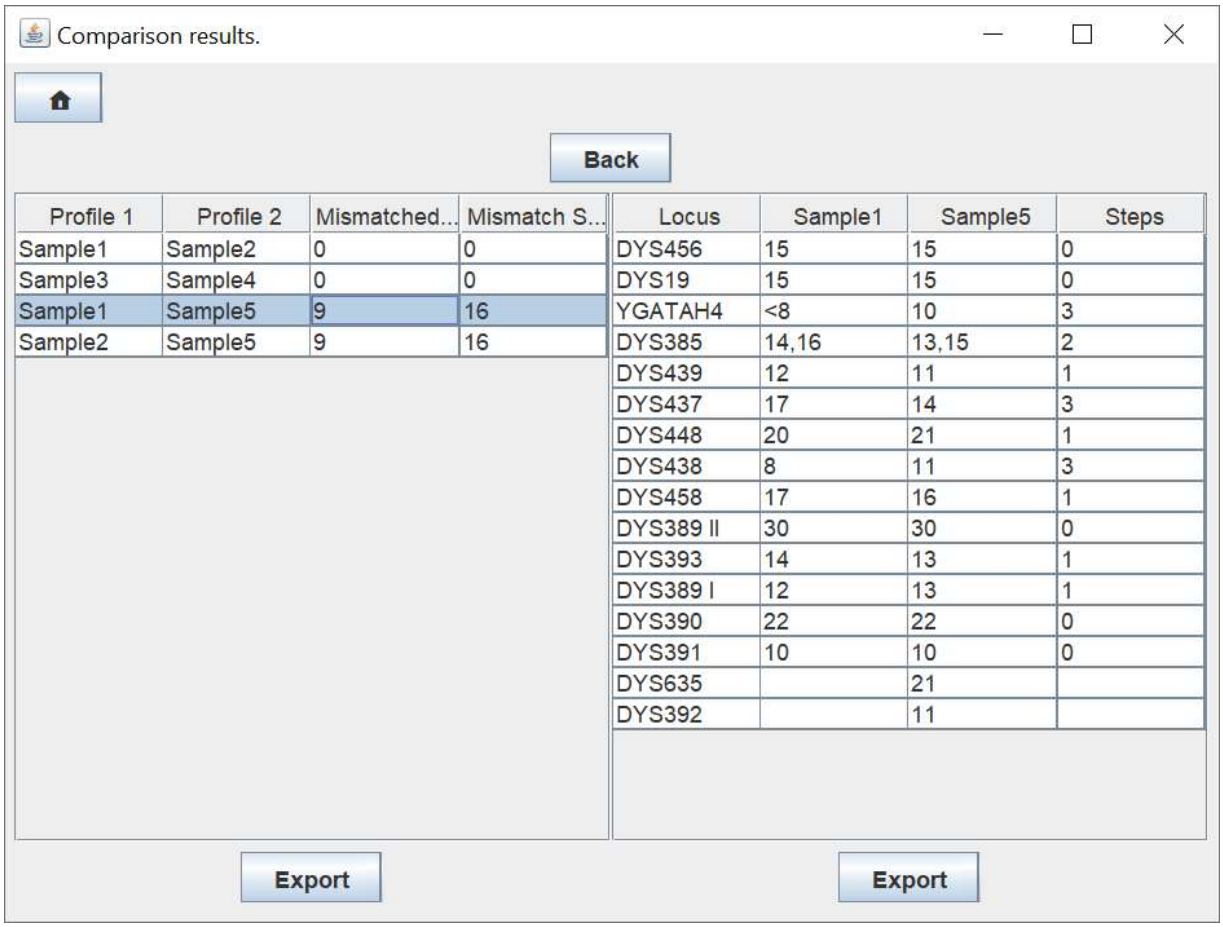


The comparison results will show all profile pairs passing the screening thresholds.

The left table shows the profile pairs with the numbers of mismatched loci and mismatched steps.

The right table shows the detailed comparison of a selected pair in the left table.

Users may export the tables as TSV (Tab-Separated Values) files.

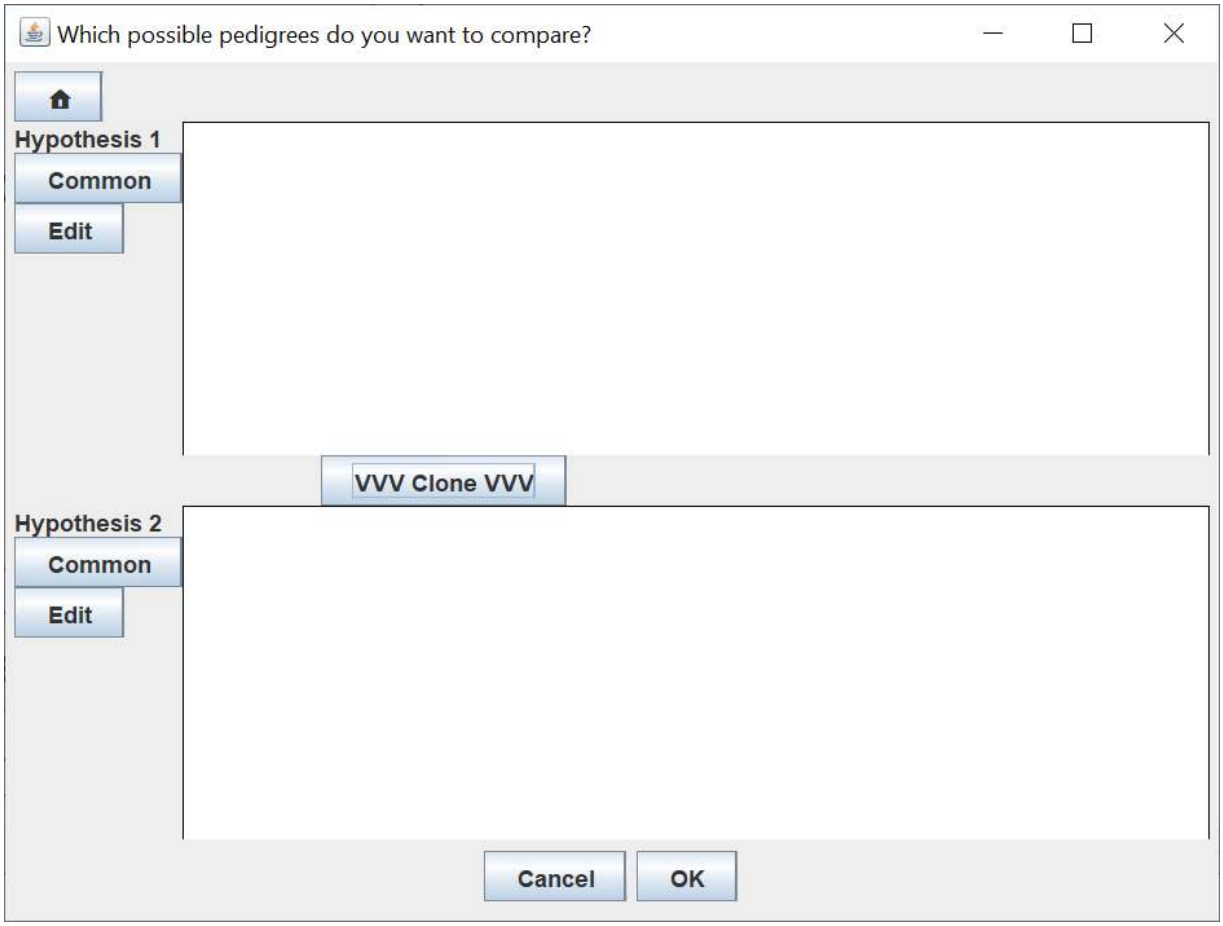


Profile 1	Profile 2	Mismatched...	Mismatch S...
Sample1	Sample2	0	0
Sample3	Sample4	0	0
Sample1	Sample5	9	16
Sample2	Sample5	9	16

Locus	Sample1	Sample5	Steps
DYS456	15	15	0
DYS19	15	15	0
YGATAH4	<8	10	3
DYS385	14,16	13,15	2
DYS439	12	11	1
DYS437	17	14	3
DYS448	20	21	1
DYS438	8	11	3
DYS458	17	16	1
DYS389 II	30	30	0
DYS393	14	13	1
DYS389 I	12	13	1
DYS390	22	22	0
DYS391	10	10	0
DYS635		21	
DYS392		11	

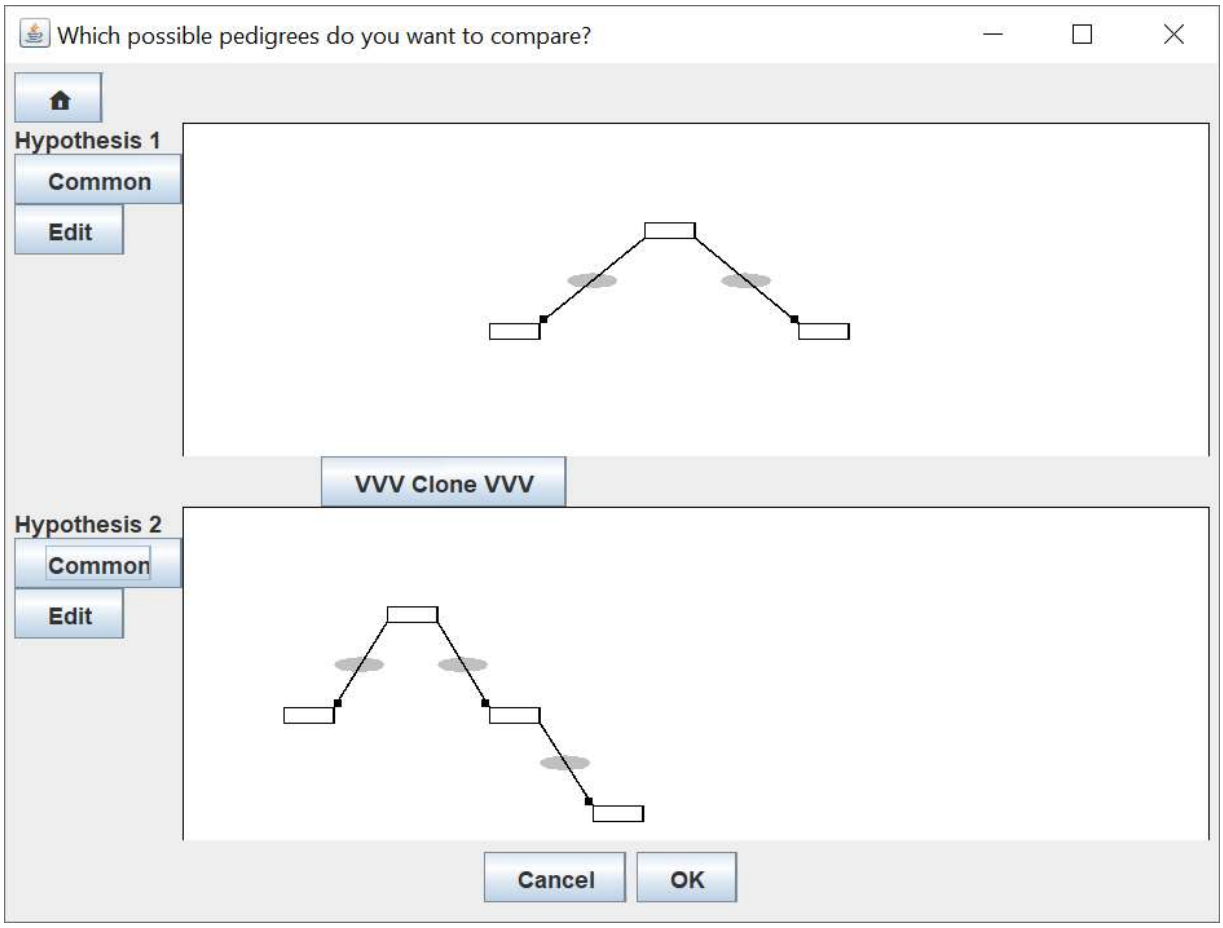
Pedigree Likelihood Ratios

MPKin-YSTR allows the users to draw two competing pedigree hypotheses, assign Y-STR haplotypes to the individuals in the pedigrees, and calculate the likelihood of each hypothesis, as well as the likelihood ratio (LR).



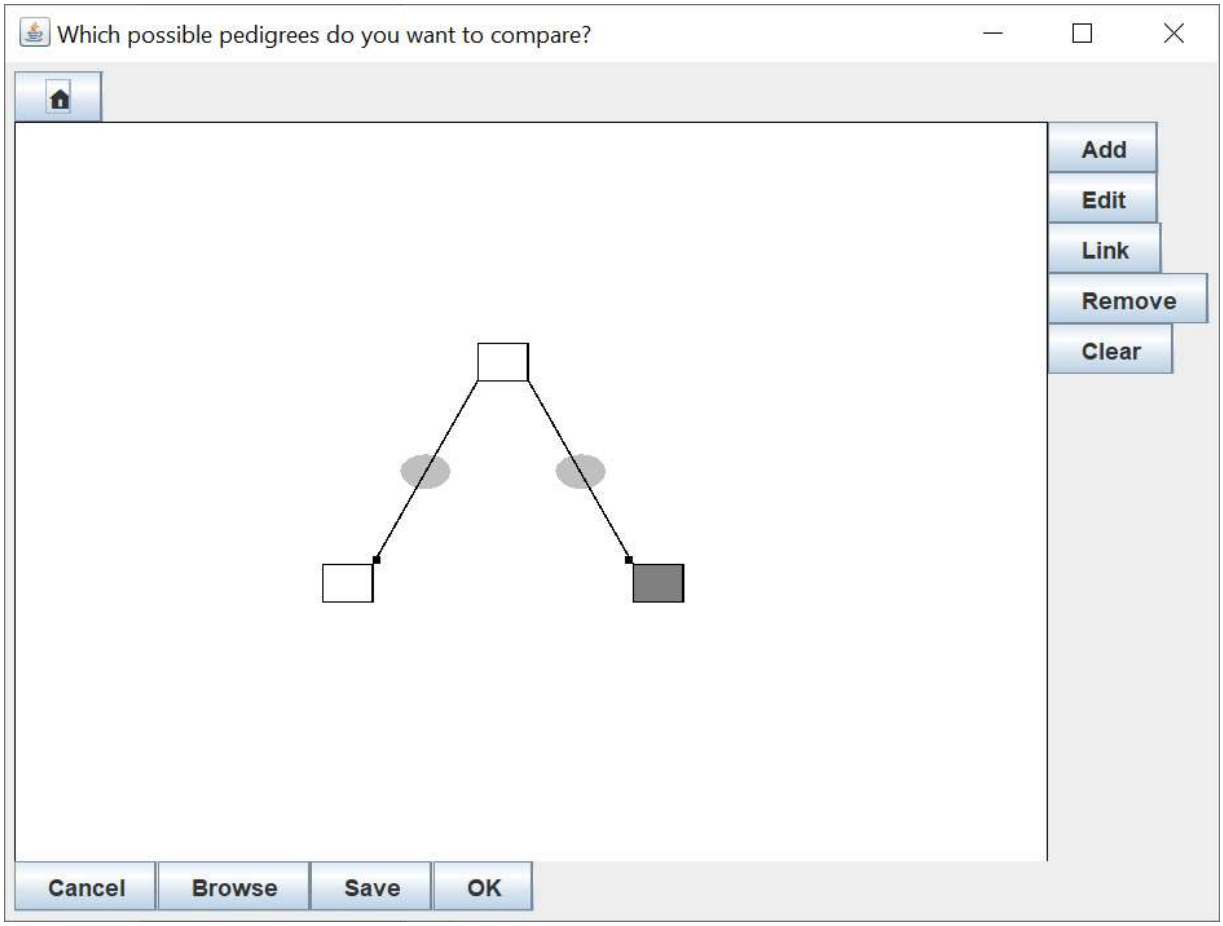
Some common pedigrees have been provided to facilitate the pedigree drawing.

An example of two competing pedigree hypotheses.

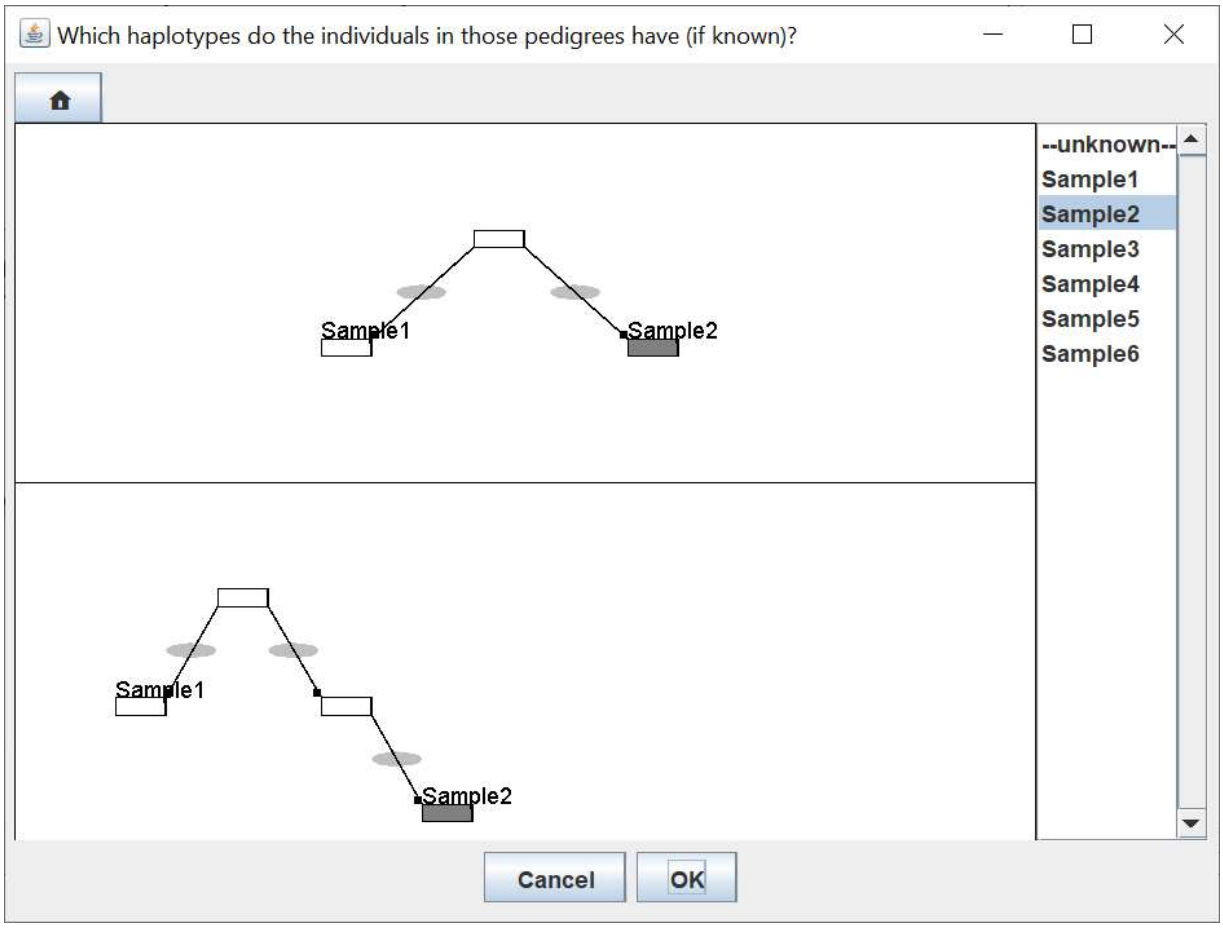


Users may define a pedigree by adding, editing, linking, or removing individual(s).

Users may also save a defined pedigree or load a saved pedigree.



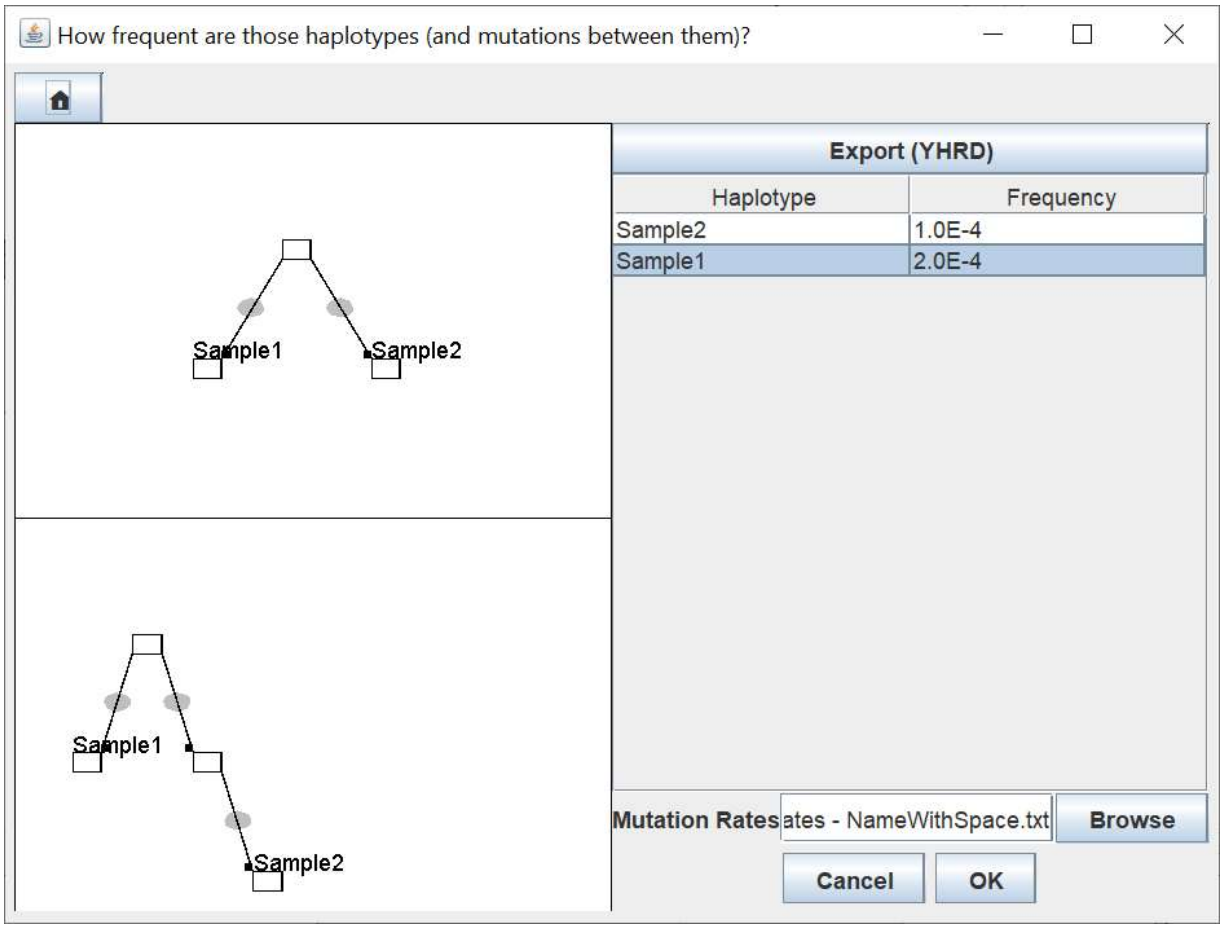
To assign the Y-STR haplotypes in the uploaded Y-STR haplotype file to the individuals in the pedigrees, the users need to click an individual first, and then click a profile on the right panel.



Users may manually enter the haplotype frequencies of all the haplotypes in the pedigrees.

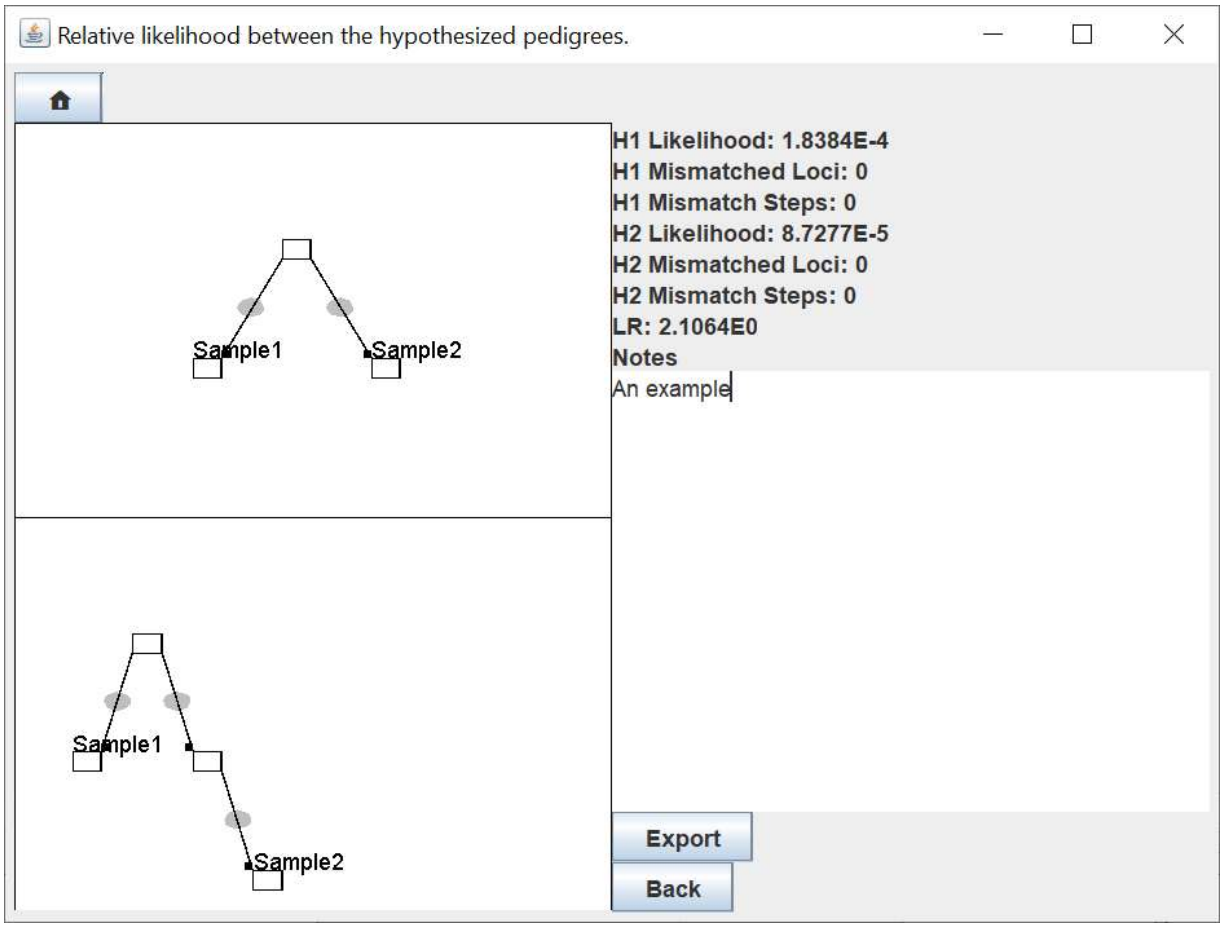
To facilitate the haplotype frequency entering, MPKin-YSTR allows the users to export all haplotypes in the pedigrees to a CSV file, which can be further uploaded to YHRD to obtain the frequencies. Since YHRD does not provide web API and multiple haplotype frequencies may be obtained in a YHRD search (for different populations, frequency calculation methods, etc.), the users have to manually enter the frequencies after obtaining the frequencies from YHRD.

Users are required to upload a Y-STR mutation rate file, which has been explained above.



MPKin-YSTR calculates the likelihoods, the number of mismatched loci, and the number of the mismatched steps for both pedigrees. In addition, the LR will be calculated.

The users may export the results of the calculation, which will include all detail information, such as the pedigrees, the Y-STR haplotypes in the pedigrees, haplotype frequencies, mutation rates, and the calculated mismatched steps of all markers, etc. In addition, the users may add notes in the exported file.

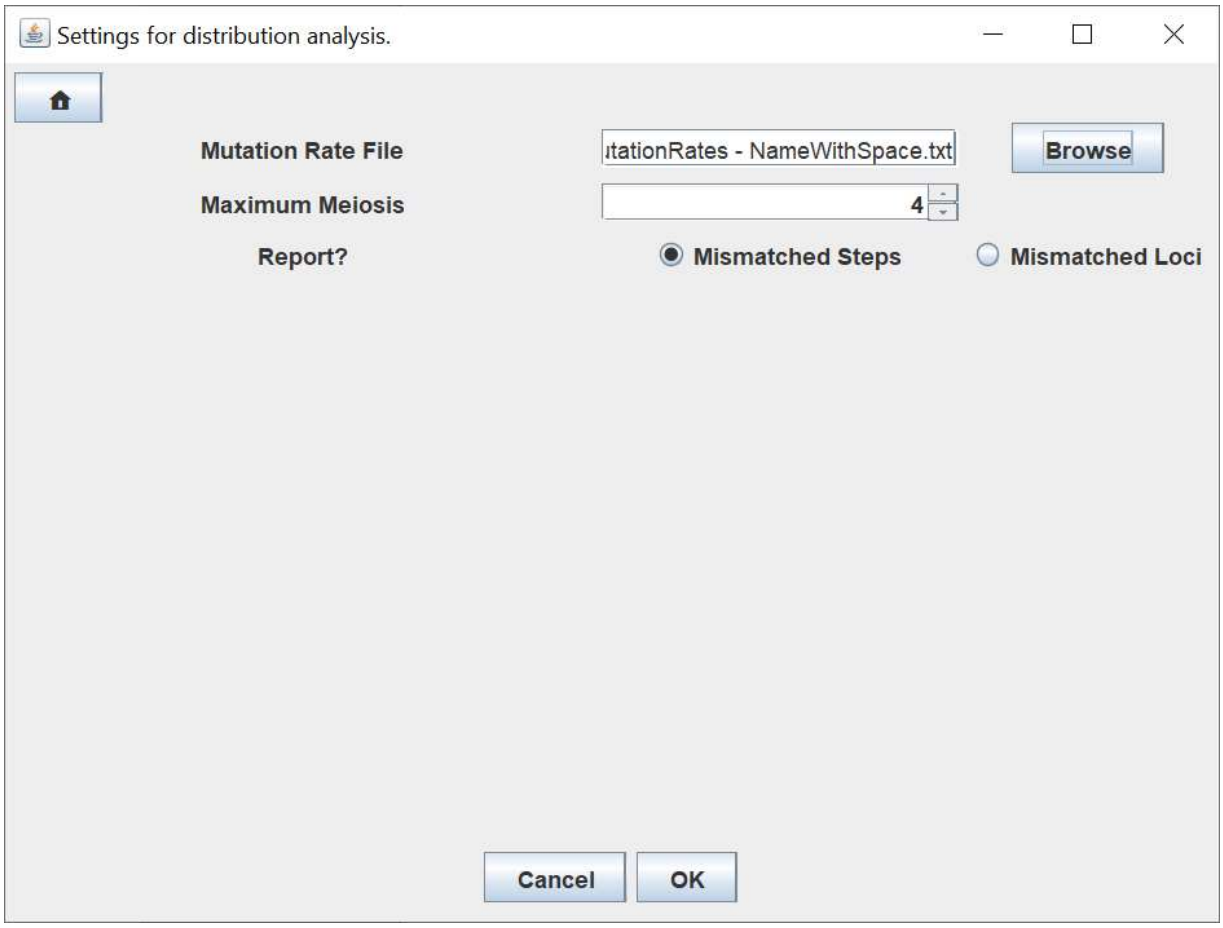


Mismatch distributions

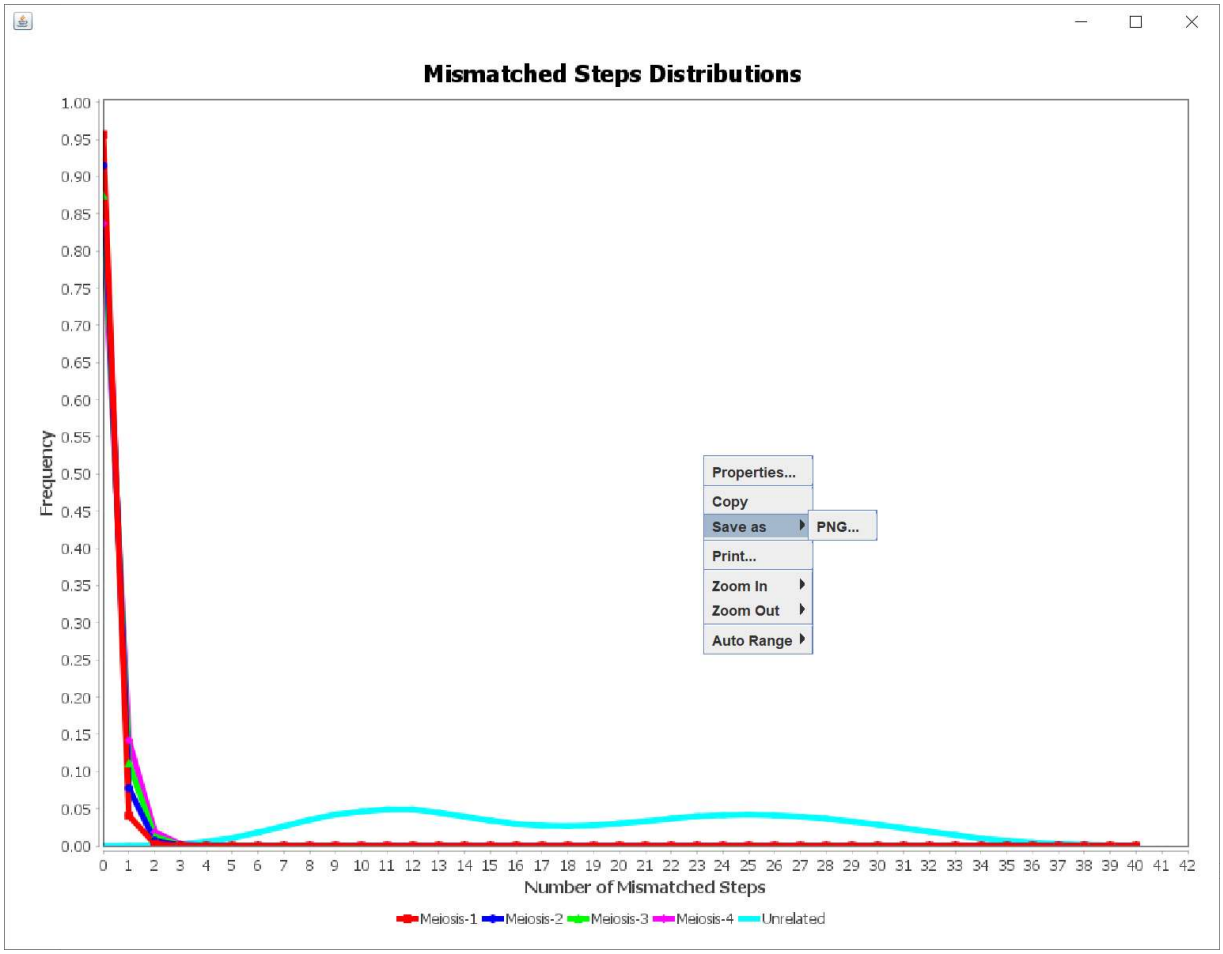
This function draws the mismatch distributions, either loci or steps, for unrelated and related pairs. Ultimately, it is to assist the DNA analysts to quickly determining if there is strong support that two profiles are from the same lineage or not.

A mutation rate file is required. The maximum meiosis also needs to be decided, which is to define the scope of a male lineage. If the Maximum Meiosis = 4, a male lineage is defined as relatives up to the first-cousin relationship.

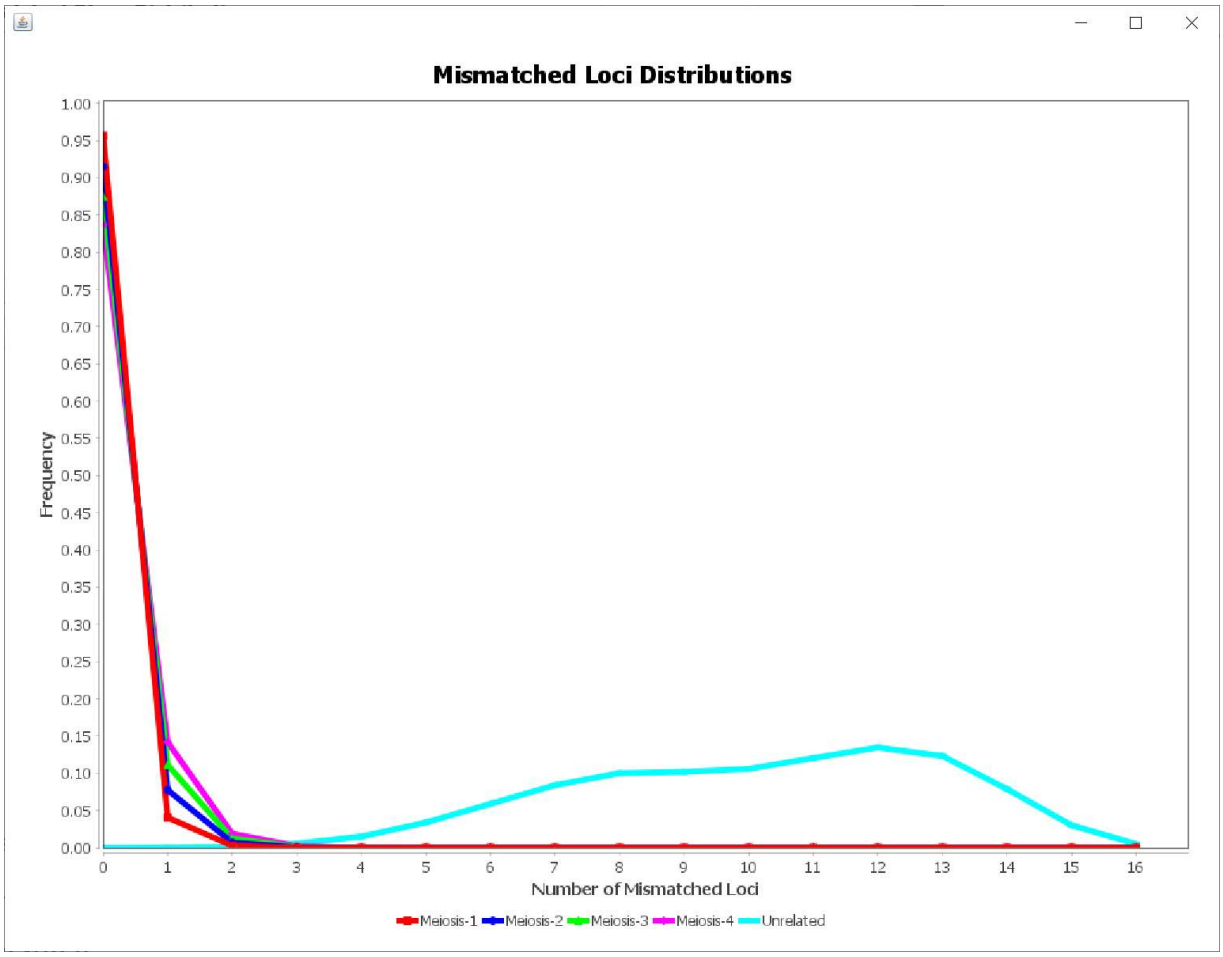
Meiosis-1 = father-son; Meiosis-2 = full-sibling; Meiosis-3 = uncle-nephew; Meiosis-4 = first-cousin.



A figure example of the mismatched steps distributions. The figure may be zoomed in or out and saved as a PNG file by clicking the right key of the mouse.

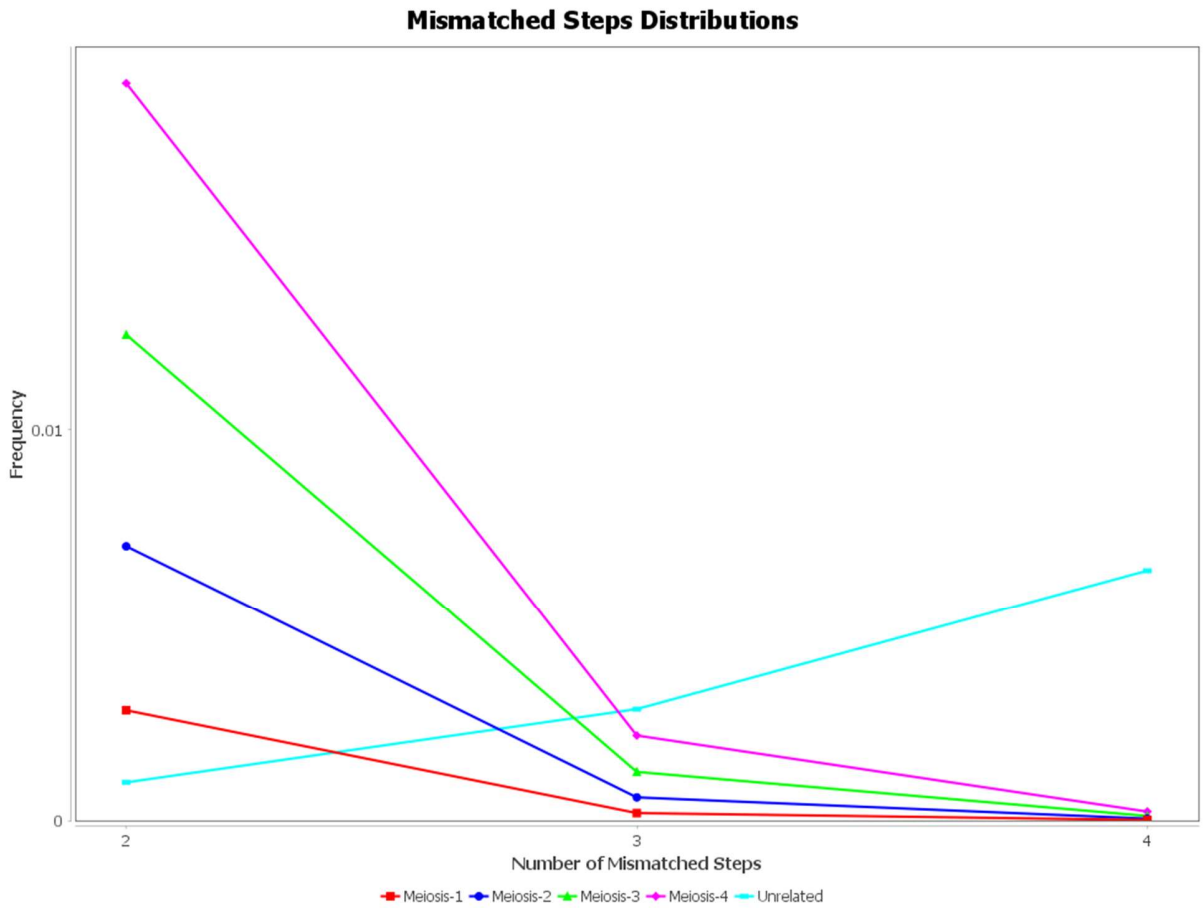


A figure example of the mismatched loci distributions.



Users may zoom in to the regions where the distribution of the unrelated pairs comes across the distributions of the related pairs, and then decide a threshold to decide paternal lineage.

In the example below, a threshold ≤2 mismatched steps would be good to conclude a male lineage.



Contacts

Questions may be sent to gejianye@gmail.com or jianye.ge@unthsc.edu.

In case a bug or error is found, please share all information (except the identifications of the test profiles) to the MPKin-YSTR authors. Thus, the authors can quickly replicate the bug or error.